

DATA ANALYSIS PROJECT REPORT

Template for Student Submission

Project Title	Seattle Weather Patterns: Exploratory Data Analysis Using Python
Student Name	SYED AL MAHMUD, GAZI MD RAFAYET HOSSEN
Student ID	[23-50168-1], [22-47836-2]
Course / Section	[A]
Instructor	[MD. TANZEEM RAHAT]
Date of Submission	[05/02/26]

1. Executive Summary

This project conducts an exploratory data analysis of the Seattle weather data using Python. The aim is to examine the changes in temperature, precipitation, and weather patterns. Pandas were used in the processing of the dataset to manipulate data, NumPy to perform numerical calculations and Matplotlib to visualize the data. The analysis shows apparent seasonal patterns, and temperature reaches its highest level in the middle months of the year, and rainfall is estimated to be concentrated on weather patterns. The findings reveal the way data analysis methods can be applied to derive valuable information out of the real-world data sets.

2. Introduction and Project Objectives

Weather data analysis plays an important role in understanding climate patterns, which can help in planning agriculture, transportation, and daily activities. This project focuses on analyzing a real-world weather dataset from Seattle.

Item	Your Content
Dataset name	Seattle Weather Dataset
Dataset source	https://raw.githubusercontent.com/vega/vega/refs/heads/main/docs/data/seattle-weather.csv
Main project goal	To analyze weather patterns and identify trends in temperature and precipitation.
Research Question 1	How does temperature vary across different months?
Research Question 2	How does precipitation differ across weather conditions?
Research Question 3 (optional)	Which weather condition occurs most frequently?

Weather data analysis plays an important role in understanding climate patterns, which can help in planning agriculture, transportation, and daily activities. This project focuses on analyzing a real-world weather dataset from Seattle.

3. Dataset Description

The dataset contains daily weather observations, where each row represents one day.

Columns :

date: Date of observation
precipitation: Amount of rainfall
temp_max: Maximum temperature of the day
temp_min: Minimum temperature of the day
wind: Wind speed
weather: Weather condition (rain, sun, fog, drizzle, snow)

3.1 Basic Dataset Information

Property	Value
Number of rows	1,461
Number of columns	6
Data types present	numerical / categorical / date-time
Target or key variable(s)	temp_max and temp_min precipitation weather
Potential issues observed at first glance	No major formatting issues, but preprocessing is required

3.2 Initial Observations

At first glance, the Seattle Weather data seems to be well-organized and clean, consisting of 1,461 records of daily data with 6 variables. The data consists of a combination of both numerical (precipitation, temperatures, and wind speed) and one categorical (weather) with five conditions. There were no missing values, which is good to analyze. The date column is now being kept as text and will need conversion to be in the form of a datetime. Moreover, there were some extreme values of precipitation and negative temperatures, which are valid, considering the climate in Seattle.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Handling Missing Values	Missing values were checked using <code>.isnull()</code>	Any missing records were removed to maintain data quality.	<pre>print(df.isnull().sum())</pre>
Convert into DateTime	The date column was converted into a datetime format using Pandas.	This allows extraction of month and year for time-based analysis.	<pre>df['date'] = pd.to_datetime(df['date'])</pre>

5. Feature Engineering

New features were developed to enhance the analysis and better address the research questions.

New Feature	How It Was Created	Why It Is Useful
Month	Extracted from the date column	Helps analyze seasonal trends in temperature and precipitation.
Year	Extracted from the date column	Helps identify long-term trends over the years.
Temp_Range	Calculated as <code>temp_max - temp_min</code>	Indicates daily temperature variation.

6. Analysis Methodology

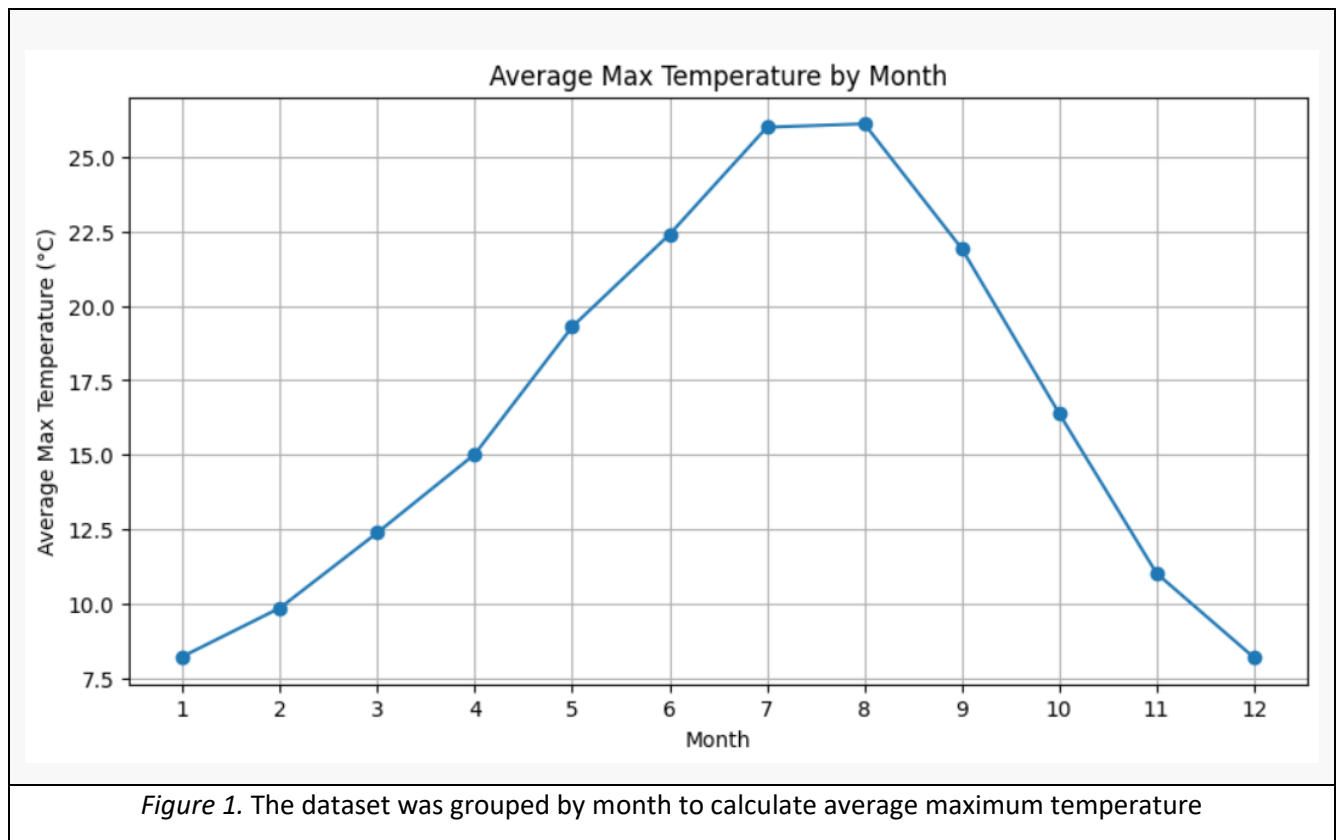
Pandas were applied to load, filter, group and aggregate data. Numbers were performed with the help of NumPy. Visualizations were generated by Matplotlib to have a better understanding.

7. Analysis and Visualizations

7.1 Research Question 1

Temperature Change: Based on the data, the data was aggregated by month to compute the average maximum temperature.

Result: There is a rise in temperature during the middle months of the year and a decline during the start and end implying seasonal change.



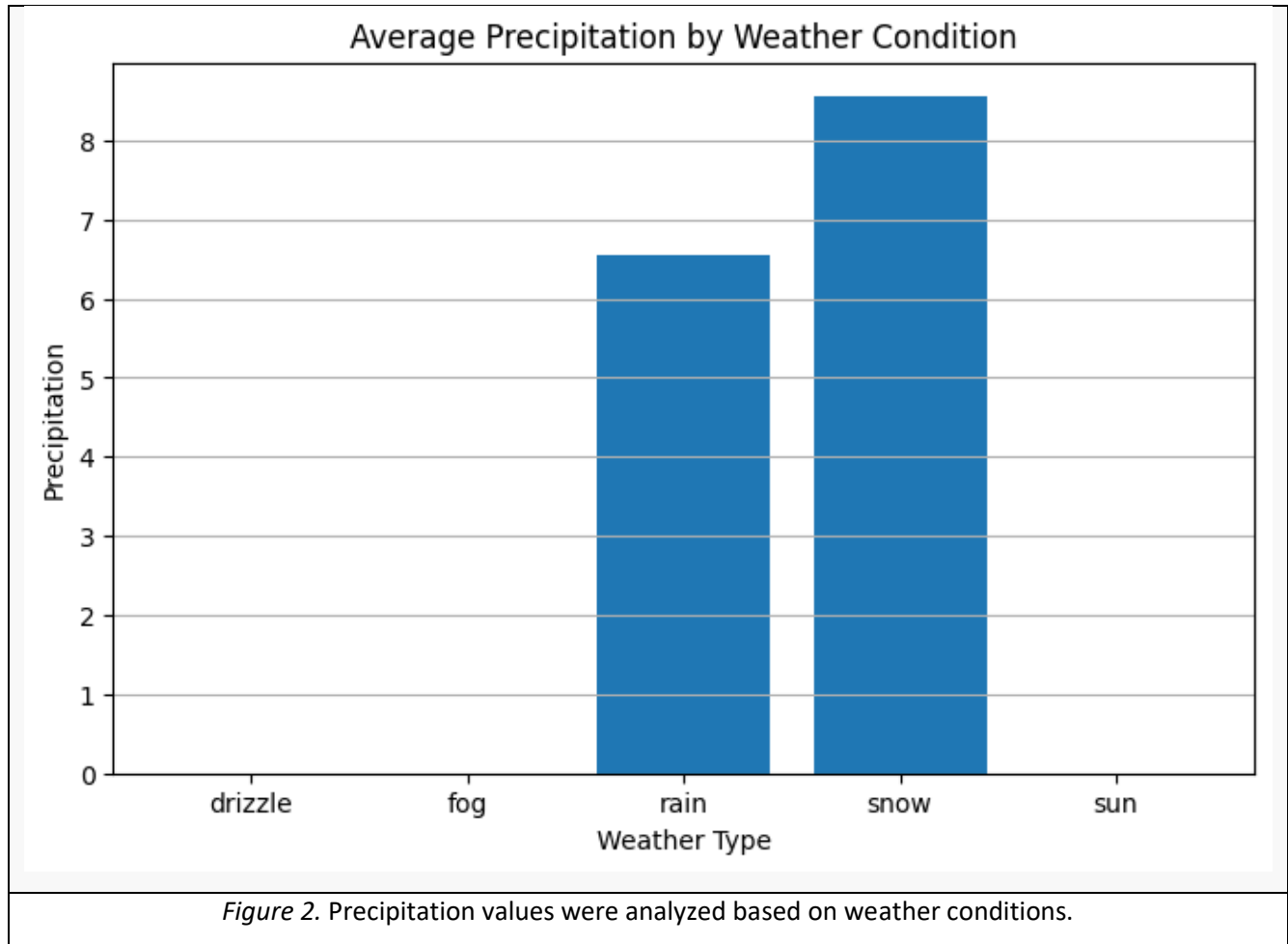
Interpretation: The line plot shows a clear seasonal trend in temperature variation across the months. Temperature gradually increases from January and reaches its peak during July and August, indicating the summer season. After this peak, the temperature declines toward December, representing the winter season.

This smooth rise and fall pattern confirm that temperature is strongly influenced by seasonal cycles. The consistency of the trend suggests that Seattle experiences predictable annual climate behavior, making temperature patterns relatively stable year to year.

7.2 Research Question 2

Precipitation Analysis: The values of precipitation were determined depending on weather conditions.

Result: Rainy weather indicates a high precipitation whereas sunny weather indicates low precipitation

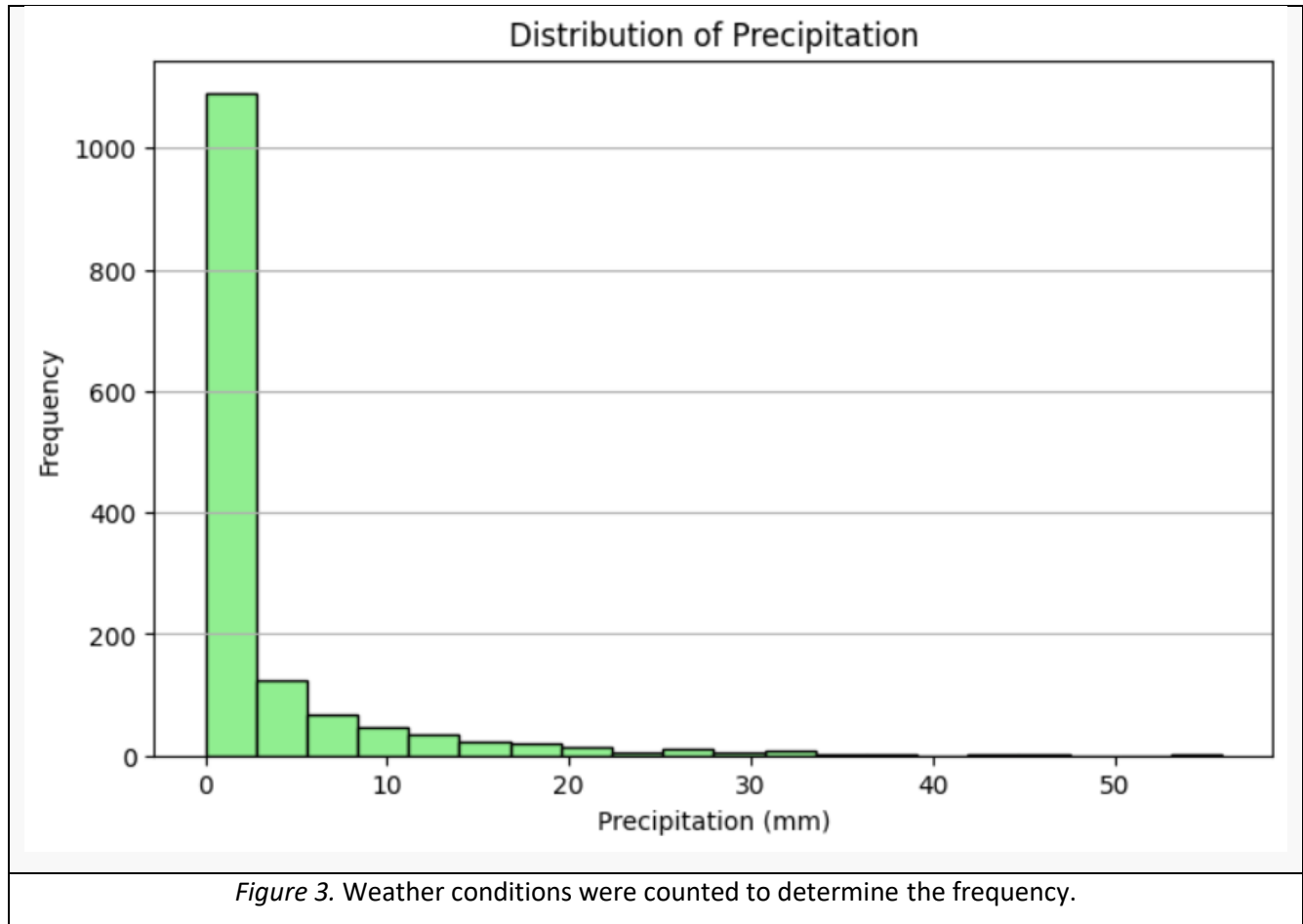


Interpretation: The chart shows that snow and rain have the highest average precipitation, with snow being slightly higher. In contrast, drizzle, fog, and sunny conditions have near-zero precipitation, indicating minimal or no rainfall. This suggests that precipitation mainly occurs during rain and snow conditions. Overall, the results confirm a strong relationship between weather type and precipitation level.

7.3 Research Question 3

Weather Distribution: The frequency was determined by counting weather conditions.

Result: Certain types of weather are much more common, which implies prevailing climate patterns.



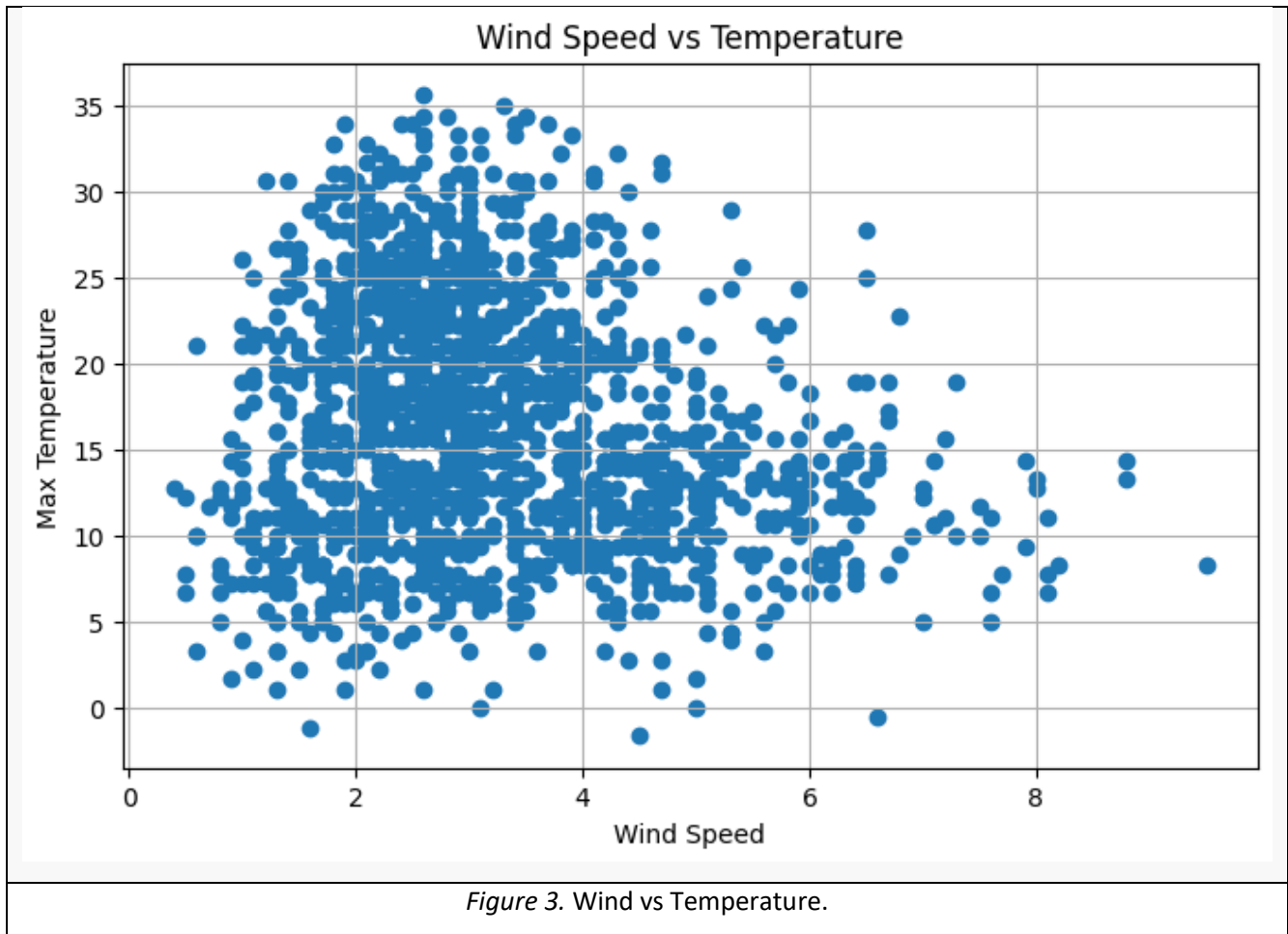
Interpretation: The histogram shows that most of the precipitation values are concentrated near 0 mm, meaning light or no rainfall occurs most frequently.

As precipitation increases, the frequency drops sharply, indicating heavy rainfall is rare in the dataset.

The distribution is highly right skewed, with a long tail extending toward higher rainfall values.

Overall, the data suggests a generally dry climate with occasional heavy rain events.

Wind vs Temperature (Scatter Plot)



Interpretation: Shows the relationship between wind speed and temperature. There is no strong pattern only weak correlation.

8. NumPy-Based Custom Computation

NumPy was used to perform statistical analysis such as calculating averages and distributions.

Example:

Mean temperature calculation

Distribution of precipitation values

Purpose: To support deeper numerical analysis beyond basic grouping.

Component	Description
Computation used	Temperature Z-score calculation and precipitation distribution analysis using NumPy
Purpose	To standardize temperature values and identify how far each value deviates from the mean, and to analyze the distribution pattern of precipitation values
Formula or logic	Z-score formula: $z = (x - \mu) / \sigma$ Where: x = value, μ = mean, σ = standard deviation. NumPy functions <code>np.mean()</code> and <code>np.std()</code> were used. For distribution, NumPy arrays were used to compute statistical summaries
Result / takeaway	The Z-score helped identify unusually high or low temperature days (outliers). The precipitation distribution showed that most values are concentrated near zero, indicating that heavy rainfall is less frequent

9. Key Findings

1. Temperature peaks during mid-year months (summer season)
2. Rainfall is concentrated in rainy weather conditions
3. Clear weather occurs most frequently
4. Temperature variation differs across seasons
5. Weather patterns follow predictable trends

10. Limitations

Dataset is limited to a single city (Seattle).

No real-time or live data is included.

Limited variables restrict deeper analysis.

Correlation does not imply causation.

11. Conclusion

This project was able to analyze the Seattle weather data with the help of Python tools. The results indicate the evident seasonal variations in temperature and precipitation. Using data cleaning, feature engineering, and visualization, significant insights were achieved. The project shows the way data analysis can be applied to comprehend the real-world patterns in an efficient manner.

12. References / Dataset Source

Dataset Source:

<https://raw.githubusercontent.com/vega/vega/refs/heads/main/docs/data/seattle-weather.csv>

Python Documentation

Pandas Documentation

NumPy Documentation